



Seasonal Variation in Chemical Composition and *in vitro* Nutrient Digestibility of Three Halophytic Shrubs (*Atriplex halimus*, *Arthrophytum scoparium* and *Tamarix africana*) from Arid Region of Algeria

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ABSTRACT

Background: Halophytic shrubs are strategic alternative forage resources for ruminants in arid and saline environments, where conventional feed availability is limited by drought, soil salinity and rangeland degradation. This study evaluated the chemical composition, *in vitro* fermentation characteristics and estimated nutritive value of three dominant halophytic shrubs from Sidi Okba, Biskra, Algeria.

Methods: Edible aerial parts of *Atriplex halimus*, *Arthrophytum scoparium* and *Tamarix africana* were collected during autumn (S1) and spring (S2). Samples were dried, ground and analysed for dry matter, mineral matter, organic matter, crude protein and crude cellulose. Organic matter digestibility was determined by *in vitro* gas fermentation and energy and protein values were estimated according to the INRA system. Data were analysed using ANOVA, Tukey's HSD test, Student's t-test and principal component analysis.

Result: Significant interspecific differences were found for most variables ($p < 0.05$). *Atriplex halimus* showed the best profile-crude protein 15.74-17.00%, moderate minerals, high digestible protein and peak energy values (0.66 UFL and 0.59 UFV/kg DM). *Tamarix africana* had the lowest profile-crude protein ~3.5%, digestibility 41.39-43.87%. No significant seasonal effect was observed ($p > 0.05$), indicating nutritional stability across periods. Principal component analysis (PCA) clearly separated the two species, with protein and energy parameters (crude protein, organic matter digestibility) as main discriminants. Overall, *Atriplex halimus* outperforms *Tamarix africana* nutritionally, with seasonal stability supporting its forage value.

Key words: *Atriplex halimus*, Halophytes, *In vitro* fermentation, Nutritional value, Ruminant feeding, Saline rangelands.

INTRODUCTION

Sustainable ruminant production in arid and semi-arid environments is increasingly constrained by land degradation, water scarcity and progressive soil salinization. These constraints reduce both the availability and nutritive quality of conventional forages and make feed shortage one of the major limiting factors for livestock systems in saline rangelands. According to Singh (2021), adaptive capacity refers the power of a system to global climate change (including climate variability and extremes) to moderate potential damages, to require advantage of opportunities, or to deal with the consequences. In this context, halophytic shrubs, which are naturally adapted to high salinity and limited water availability, have attracted attention as resilient alternative forage resources (Hasnain *et al.*, 2023). Among these shrubs, *Atriplex halimus* is widely used by small ruminants in Mediterranean and North African drylands because of its high tolerance to salt stress and its capacity to maintain green biomass even under harsh conditions. Additionally, they are a source of pharmaceutical compounds, raw materials for bio carburants and may be used in phytoremediation (Mahmoudi *et al.*, 2023).

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The feeding value of halophytes depends on their chemical composition, digestibility and ruminal fermentation behavior. *In vitro* gas production techniques provide a standardized approach for evaluating fermentation kinetics and organic matter digestibility and can help estimate the contribution of these plants to ruminant diets (Haddi *et al.*, 2003; Riasi *et al.*, 2012). However, the nutritive characteristics of halophytes are not constant. Seasonal variation may modify protein concentration, fiber accumulation, vary mineral content and digestibility through changes in plant maturity, water availability and soil salinity (Cooke *et al.*, 2024; Oliveira *et al.*, 2025).

Despite increasing interest in saline forage resources, comparative studies on co-occurring halophytic shrubs under the same ecological conditions remain limited in North African arid ecosystems. The Sidi Okba region of Biskra, Algeria, is strongly affected by salinization and therefore provides a relevant natural setting for studying the nutritional adaptability of local halophytes. Previous work in this area reported saline to very saline soils, with electrical conductivity values that may reach approximately 3.90 dS/m (Bekaddour *et al.*, 2018).

Focusing on three key halophytic shrubs, *Atriplex halimus*, *Arthrophytum scoparium* and *Tamarix Africana*, from the saline rangelands of Sidi Okba, this study assessed seasonal variations in chemical composition and *in vitro* organic matter digestibility using rumen liquor. Multivariate statistical methods were further applied to establish relationships among nutritional parameters and support sustainable feeding strategies for ruminants in arid regions.

MATERIALS AND METHODS

Study area

The study was conducted in Sidi Okba (Biskra, Algeria; 34°49'N, 5°55'E, elevation 110 m), an arid environment located on the edge of the Sahara. The region is characterized by an average annual temperature of 22°C and low annual precipitation (120 to 150 mm; ONM, 2023). According to Rechachi (2017), the local sandy-loam soils exhibit marked salinity (EC: 1.5 to 6 dS/m) which favors the development of halophilic flora dominated by *Atriplex halimus*, *Arthrophytum scoparium* and *Tamarix africana*. Under these arid and saline conditions, where sensitive plants cannot survive, these tolerant shrubs provide a vital source of consumable biomass for livestock feed (El-Shaer, 2010).

Plant material and sampling design

Forage samples from three dominant halophytic shrub species (*Atriplex halimus*, *Arthrophytum scoparium* and *Tamarix africana*) were collected at Sidi Okba during the fall (S1) and spring (S2) seasons of 2020-2021. For each species, the edible parts (leaves and young twigs measuring 15-25 cm) were manually collected from five

adult plants within 50 × 50 m plots. The samples were pre-dried in the shade, then oven-dried at 60°C for 48 hours. After drying to constant weight, the plant material was ground to 1 mm and stored in airtight vials for physicochemical and *in vitro* fermentation analyses, in accordance with standardized protocols (AOAC, 2019; Medjekal and Bousseboua, 2016).

Chemical analysis

The present study was carried out at the Biosystematics Laboratory, CRSTRA (Biskra, Algeria) during 2023-2024. Dry matter (DM, 105°C/24 h), ash/minerals (550°C/6 h), crude protein (CP, Kjeldahl method; ISO, 1997) and crude cellulose (BC) were determined following standard procedures (AOAC, 2019). Organic matter (OM) was calculated as the difference between DM and ash.

In vitro fermentation and digestibility

Organic matter digestibility (OMD) was determined by *in vitro* fermentation according to the VDLUFA 25.1 protocol of the Association of German Agricultural Analytic and Research Institutes (VDLUFA, 2012). The buffer and nutrient solutions were prepared following Menke and Steingass (1988). Approximately 200 mg of dry matter were weighed into 100 mL glass fermentation syringes as described by Sommart *et al.* (2000).

Rumen inoculum was obtained from two cattle slaughtered at the local abattoir, filtered and kept warm before use following Salem (2005) and Mould *et al.* (2005). Each syringe received a mixture composed of one-third rumen inoculum and two-thirds mineral-buffered solution. Incubations were carried out under anaerobic conditions at 39°C to simulate ruminal conditions. All treatments were performed in three independent replicates. Gas produced after 24 h was corrected using blanks and standards before estimating digestibility.

Estimation of nutritive values

The organic matter digestibility (OMD, %) values obtained using the *in vitro* fermentation method (gas production) were used to estimate the nutritional value of the plants according to the INRA 2018 system (Guamán-Rivera *et al.*, 2023). The energy parameters, milk feed unit (UFL) and meat feed unit (UFV) as well as the protein parameters dietary protein digestible in the intestine (PDIA), microbial protein digestible in the intestine (PDIM), total PDI and rumen protein balance (RPB), were calculated using the INRA prediction equations (Chenost *et al.*, 2001). This approach enabled a reliable assessment of the energy and nitrogen contributions of the studied halophytic species for ruminant nutrition.

Statistical analysis

Statistical analyses were conducted using R software. Data normality was confirmed via the Shapiro-Wilk test, Q-Q plots and histograms ($p > 0.05$). One-way ANOVA followed by Tukey's HSD test evaluated species differences, while

Student's t-test assessed seasonal variation (S1 vs. S2). Principal component analysis (PCA) was performed to explore multivariate relationships between species and nutritional parameters ($p < 0.05$).

RESULTS AND DISCUSSION

Chemical composition

Chemical profiles differed by species but remained stable across seasons ($p > 0.05$). Dry matter was high (93.29-94.78%). *Atriplex halimus* had the highest crude protein (15.74-17.00%), high organic matter (up to 79.32%) and low crude fiber (Table 1). Conversely, *Arthrophytum scoparium* accumulated up to 57.34% ash with lowest organic matter (37.44%), while *Tamarix africana* showed minimal protein (3.45-3.57%).

Estimated nutritive values

The analysis of estimated nutritive variables showed significant differences among species (Table 2). Digestible organic matter differed significantly ($F = 7.771$; $p = 0.0134$). Energy variables showed stronger differences,

with highly significant effects for UFL ($F = 24.34$; $p = 0.0006$) and UFV ($F = 32.91$; $p = 0.0003$). Protein-related variables also differed, particularly PDIA, PDIM and PDI (Table 4). These results indicate marked variability in energy and digestible protein supply among the evaluated halophytes (Fig 2).

Digestibility and energy values

Determination of digestibility of nutrients is an important criterion for quality evaluation and feeding value (Barman *et al.*, 2023). Organic matter digestibility and energy values differed significantly among species (Table 3). *Arthrophytum scoparium* S2 recorded the highest OMD value (67.31%), whereas *Tamarix africana* showed the lowest digestibility values (41.39-43.87%). *Atriplex halimus* showed intermediate to high OMD values, with 60.55% during S1 and 48.34% during S2. The highest energy values were recorded in *A. halimus* S1, which reached 0.66 UFL/kg DM and 0.5858 UFV/kg DM (Table 4). In contrast, *T. africana* had the lowest energy density and the lowest digestible protein values.

Table 1: Chemical composition ((% DM) of three halophytic forage species harvested during two contrasting seasons.

Nutrient (%)	<i>A. halimus</i> S1	<i>A. halimus</i> S2	<i>A. scoparium</i> S1	<i>A. scoparium</i> S2	<i>T. africana</i> S1	<i>T. africana</i> S2
DM	94.56±0.14	94.44±0.11	94.58±0.02	94.78±0.02	93.29±0.01	93.30±0.05
MM	21.16±1.78	15.12±0.18	24.40±1.77	57.34±13.80	35.50±14.28	39.30±17.40
OM	73.40±1.03	79.32±0.14	70.18±0.89	37.44±6.91	57.79±7.14	54.00±8.72
CP	17.00±3.54	15.74±3.68	12.70±1.35	12.42±1.53	3.45±2.61	3.57±2.57
BC	8.26±1.15	9.03±1.16	25.62±1.56	25.36±1.60	17.00±0.45	19.48±1.23

DM: Dry matter; MM: Mineral matter/ash; OM: Organic matter; CP: Crude protein; BC: Crude cellulose; S1: Autumn season; S2: Spring season.

Table 2: Analysis of variance for digestibility, energy and protein variables across species.

Nutritional profile	F-value	p-value	Significance
dOM (%)	7.771	0.0134	*
UFL (UF/kg DM)	24.34	0.0006	***
UFV (UF/kg DM)	32.91	0.0003	***
PDIA (g/kg DM)	10.37	0.0065	**
PDIM (g/kg DM)	10.86	0.0058	**
PDI (g/kg DM)	16.12	0.0020	**
BPR (g/kg DM)	7.664	0.0139	*

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. dOM: Digestible organic matter; BPR: Ruminant protein balance.

Table 3: Comparison of organic matter digestibility among the studied species and seasons.

Group	Species/season	OMD (%)
1	<i>Arthrophytum scoparium</i> S2	67.31 ^a
2	<i>Atriplex halimus</i> S1	60.55 ^{ab}
3	<i>Arthrophytum scoparium</i> S1	51.23 ^{ab}
4	<i>Atriplex halimus</i> S2	48.34 ^{ab}
5	<i>Tamarix africana</i> S2	43.87 ^b
6	<i>Tamarix africana</i> S1	41.39 ^b

Means followed by different letters within the same column differ significantly at $p < 0.05$.

Principal component analysis

PCA showed that the first two components explained most of the variability in the nutritive data. PC1 accounted for 84.46% of the variance and was mainly associated with protein and energy efficiency. Samples of *Atriplex halimus* were positioned toward higher digestible protein and energy values, whereas *Tamarix africana* samples were separated by lower nutritive performance. PC2 explained 13.15% of the variance and was primarily associated with organic matter digestibility, particularly for *Arthrophytum scoparium* S2. The PCA biplot (Fig 1) confirmed the nutritional separation of the species and highlighted the superior overall profile of *A. halimus*.

Sensitivity measures the degree to which a system (whether socio-economic or ecological) is affected by environmental stress, while adaptive capacity represents its ability to adjust to climate change in order to mitigate damages, manage consequences, or take advantage of new opportunities.

This study demonstrates that the nutritive value of Sidi Okba (Biskra, Algeria) saline rangeland halophytes is strongly species-dependent, differing in mineral matter, crude protein, crude cellulose, digestibility and energy-protein supply. These variations reflect distinct ecological strategies, showing that halophytes are not a homogeneous forage group. Since forage nutritional value relies primarily on energy value (organic matter digestibility) and voluntary dry matter intake (Sidi *et al.*, 2016), these findings highlight

the nutritional dynamics and *in vitro* digestibility of these three shrubs as alternative feed resources for arid environments. According to Chehma *et al.* (2010), forage value is directly related to the volume of intake. The volume of intake is strongly influenced by the digestibility of the plant and adjusted by metabolic processes according to the animal's nutritional requirements.

The results revealed substantial interspecific variability, notably in mineral content, crude protein and organic matter digestibility, reflecting distinct nutritional profiles and ecological adaptations among the studied species. According to Brinis and Belkhodja (2015), phytomass yield of *Atriplex halimus* is a function of the severity of salt stress. This parameter, due to its direct use as livestock feed, becomes a major determining factor for herders.

Atriplex halimus showed the most favorable overall nutritive profile. Its crude protein content remained above 15% in both seasons, while crude cellulose was comparatively low. This combination is favorable for ruminant feeding, particularly in arid zones where nitrogen availability is often limited. The results support previous findings describing *Atriplex* species as salt-tolerant shrubs with a relatively high protein content and a useful role in degraded rangelands (Hasnain *et al.*, 2023). The relative stability of *A. halimus* between seasons also suggests a degree of metabolic resilience under fluctuating environmental conditions. *Atriplex halimus* demonstrated a consistently superior nutritive value, with crude protein levels exceeding 15% and moderate fiber content, compared to *Arthrophytum scoparium* and *Tamarix africana*. These findings corroborate previous reports by Hasnain *et al.* (2023) and Wang *et al.* (2022), which emphasized the salt tolerance and protein-rich nature of *Atriplex* species in degraded rangelands. *Atriplex* is a keystone species in grazing areas, whose ability to survive in these challenging environments is due to internal endogenous adaptations (Benaradj *et al.*, 2014). The stability of the chemical composition and fermentation traits of *A. halimus* across seasons, as observed in this study, further suggests its metabolic resilience under fluctuating environmental conditions, aligning with Cooke *et al.* (2024).

In contrast, *A. scoparium* showed pronounced seasonal variability in ash content, particularly during the spring/dry season ($57.34 \pm 13.8\%$) given in (Table 1),

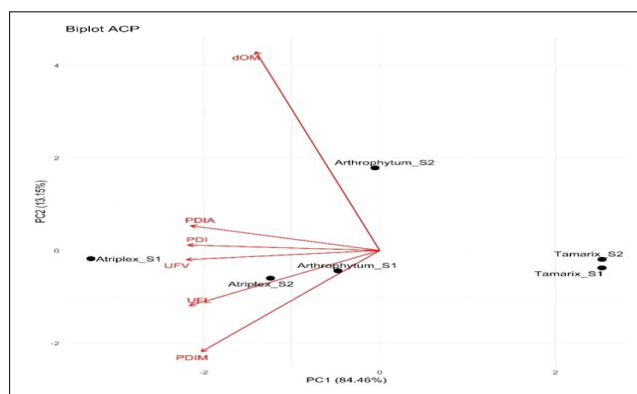


Fig 1: Principal component analysis (PCA).

Table 4: Estimated protein and energy profile of the studied forage species.

Group	Species/season	PDIA	PDIM	PDI	UFL	UFV
1	<i>A. halimus</i> (S1)	64.46 ^a	43.14 ^a	107.60 ^a	0.66 ^a	0.5858 ^a
2	<i>A. halimus</i> (S2)	57.65 ^a	40.03 ^a	97.67 ^a	0.53 ^{ab}	0.4243 ^b
3	<i>A. scoparium</i> (S2)	46.33 ^{ab}	35.39 ^{ab}	81.73 ^{ab}	0.51 ^b	0.4232 ^b
4	<i>A. scoparium</i> (S1)	39.47 ^{ab}	40.04 ^{ab}	79.50 ^{ab}	0.40 ^{bc}	0.3725 ^{bc}
5	<i>T. africana</i> (S1)	20.94 ^b	34.95 ^b	55.88 ^b	0.32 ^c	0.2449 ^d
6	<i>T. africana</i> (S2)	18.43 ^b	35.03 ^b	53.46 ^b	0.32 ^c	0.2525 ^{cd}

PDIA: Digestible protein of dietary origin; PDIM: Digestible microbial protein; PDI: Total digestible protein in the intestine; UFL: Forage unit for milk; UFV: Forage unit for meat. Values with different letters in the same column differ significantly at $p < 0.05$.

indicating enhanced salt accumulation under water stress (Mansour and Ali, 2021). Similar elevated total mineral matter values were obtained by Zirmi-Zembri and Kadi (2016) for *Arthrophytum scoparium* and by Haddi *et al.* (2009) for *Tamarix* and *Atriplex* in arid zones. While this trait underpins survivability in saline soils, the excessive mineral content may present feed management challenges for ruminants, warranting balanced inclusion in diets to avoid mineral overload.

The high mineral matter recorded in *Arthrophytum scoparium*, especially during spring, indicates marked salt accumulation. This trait may support survival under saline and drought-prone conditions, but it can reduce organic matter concentration and may limit direct use in diets if the mineral load is excessive. Therefore, practical incorporation of *A. scoparium* should be managed carefully, ideally by combining it with low-salt forages or concentrates to reduce the risk of mineral imbalance in grazing animals.

Crude protein is a key parameter of forage nutritional value, essential for tissue development and animal performance (Mehta *et al.*, 2025). Compared to *Atriplex halimus*, *Tamarix africana* exhibits lower crude protein content and organic matter digestibility. Although its role as a primary forage source is limited, *T. africana* remains vital to the saline rangeland ecosystem for soil stabilization and vegetation cover, thereby serving as a supplementary food resource.

As stated by Attia-Ismail (2018), the nutritional potential of halophytic species is strictly determined by their level of physiological maturity. Kessler (1990) asserts that *Atriplex halimus* is a particularly useful forage plant in summer and autumn because of its high protein and mineral content, reducing the shortage of food resources before spring. The plant has the highest energy value during the flowering phase and its protein concentration is highest when the grain is mature. According to Yerou *et al.* (2025), the study of *Atriplex halimus* in marginalized saline soils of northwestern Algeria shows that these halophytic species possess significant potential in terms of productivity and nutritional value. Natural *Atriplex* production can reach 2 to 3 tons of dry matter (DM) per

hectare, corresponding to approximately 1000 to 1500 feed units (FU) per hectare.

Tamarix africana is characterized by low crude protein ($\approx 3.5\%$), low dry matter digestibility (41.39-43.87%) and high cellulose levels, limiting its feeding value relative to the other species. This observation is consistent with Haddi *et al.* (2009) and Ben Rejeb *et al.* (2020), who documented reduced digestibility in *Tamarix spp.* The low *in vitro* fermentation parameters and reduced intestinally digestible protein (PDIA, PDI) observed reinforce its limited potential as a primary forage but highlight its ecological role in soil stabilization.

In vitro nutrient digestibility results, confirm the nutritional advantage of *A. halimus*, with higher UFL, UFV, PDIA and PDI values (Table 4), reflecting superior energy and protein availability. This aligns with the work of Meradi *et al.* (2024) regarding the contribution of *Atriplex* to the protein requirements of ruminants in steppe and saline environments. *Atriplex halimus* appears to be a significant contributor to meeting ruminant digestible protein requirements (PDIA, PDI), as it exhibits a generally favorable and positive ruminal protein balance (RPB) (Fig 2). Our results are consistent with these findings. By comparing Badarou *et al.* (2020) results with our results, we observe that plants palatable to ruminants and with a high milk feed unit also have a high meat feed unit (UFV) and high nitrogen digestibility. This is indeed the case for *Atriplex halimus* in autumn and *Arthrophytum scoparium* in spring, with digestibility rates of 60.55% and 67.31% respectively. One limitation of this study is the reliance on *in vitro* analyses, which, while informative, may not fully replicate live animal intake.

The absence of a significant seasonal effect is an important practical finding: it suggests that the halophytes studied, particularly *A. halimus* provide a relatively stable year-round forage resource, which is an asset for feed planning in arid zones where annual herbaceous plants fluctuate widely. However, these results must be interpreted within the limitations of the two sampling periods and local ecological conditions.

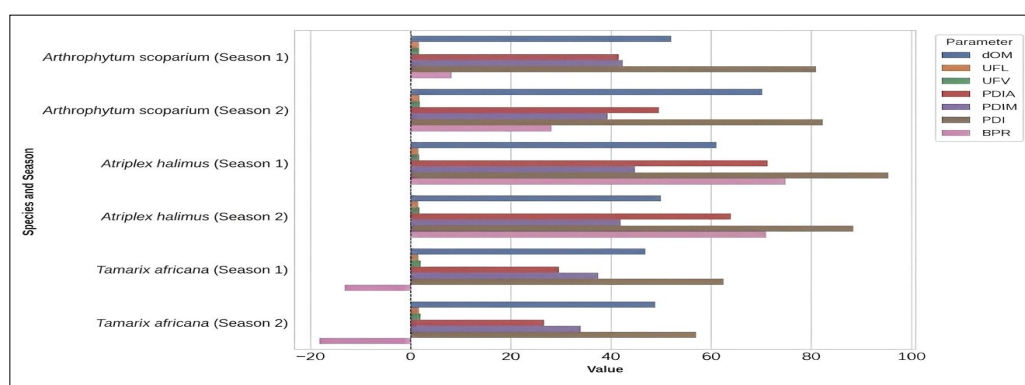


Fig 2: Nutritional parameters by species and season.

The principal component analysis (PCA) confirms the nutritional distinction between the three halophytes: *Atriplex halimus* is characterized by high energy value and digestible protein content and *Tamarix africana* shows lower performance, whereas *Arthrophytum scoparium* occupies an intermediate position, combining good digestibility (at S2) with high mineral accumulation (Fig 1). However, as the study relies on “*in vitro*” measurements, it has limitations, as it does not replicate actual animal ingestion, selectivity, or metabolism. “*In vivo*” trials and ingestion studies are required to determine safe inclusion rates and optimize the use of these forages for small ruminants in saline areas.

CONCLUSION

This study provides evidence that the three halophytic shrubs evaluated in the saline rangelands of Sidi Okba differ substantially in their nutritive value. *Atriplex halimus* was the most promising species because it combined high crude protein content, favorable digestible protein values, higher energy supply and relative stability across seasons. *Arthrophytum scoparium* showed useful digestibility in spring but also high mineral accumulation, while *Tamarix africana* had the lowest protein and digestibility values.

The findings support the integration of *A. halimus* into feeding strategies for small ruminants in arid and saline environments. However, *in vivo* validation is required before broad practical recommendations can be made. Future studies should evaluate intake, animal performance, health responses and optimal combinations of halophytes with conventional feed resources. Integrating suitable halophytes into agro-pastoral systems may improve forage availability and livestock resilience in marginal saline areas of North Africa.

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Disclaimers

The views belong solely to the authors, not their institutions. Authors strive for accuracy but accept no liability for any losses caused by using this content.

Conflict of interest

The authors declare that there are no conflicts of interest regarding the publication of this article. No funding or sponsorship influenced the design of the study, data collection, analysis, decision to publish, or preparation of the manuscript.

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